

Cultura Viva

An on-device multimodal audio guide for Gaudí’s Barcelona
Model selection, benchmarks, and bill of materials

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Abstract

Cultura Viva is an interactive multimodal AI audio guide for Antoni Gaudí’s monuments in Barcelona — Sagrada Família, Park Güell, Casa Milà (La Pedrera) and Casa Batlló — running *entirely on-device* on an Arduino UNO Q. A visitor photographs an architectural element, asks a question aloud, and hears a spoken answer in the guide personality they picked. No network round-trip is involved at inference time.

This report documents the engineering work behind the device: the four-stage pipeline (vision $\rightarrow$ speech-to-text $\rightarrow$ knowledge retrieval + small language model $\rightarrow$ speech synthesis), the comparative benchmarks that selected each model under the board’s memory and latency budget, the latency optimisations that make the interaction feel immediate, and the bill of materials with per-component prices.

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1 System overview

1.1 Target platform

The whole system runs on a single Arduino UNO Q, which is a dual-brain board:

- **STM32 microcontroller (MCU).** Real-time C++ Arduino sketch driving the ST7735S LCD, the physical inputs (mode switch, action button, Modulino buttons/knob), the Modulino buzzer, GPS ingestion and minimap rendering.
- **Qualcomm QRB2210 Linux processor (MPU).** Quad-core ARM Cortex-A53 @ 2.0 GHz, LPDDR4 RAM, 32 GB eMMC, Debian Linux. Runs the Python AI pipeline: ONNX vision classification, faster-whisper STT, knowledge-graph retrieval, a quantised small language model via `llama-cpp-python`, and Piper TTS.

The two halves talk over Arduino Bridge RPC. The board variant used for all measurements reports 3667 MB of usable RAM, i.e. the 4 GB model.

1.2 The pipeline

Figure 1 shows the full interaction. Two things about its shape are worth stating up front, because they are easy to get wrong from a prose description.

- **The guide personality is chosen before any photo is taken.** It is a setup step, not a step in the answering loop: the visitor picks Artistic, Technical or Child on the Modulino buttons, and that choice then governs both the system prompt handed to the SLM and the Piper voice used for playback. Because the personality is already fixed when a photo is confirmed, the whole stable prompt prefix — personality instructions plus the recognised element’s fact sheet — is known before the visitor has said a word, which is exactly what makes the prefill overlap in §3.4 possible.
- **The GPS does one job only: it picks which minimap is shown.** The NEO-6M feeds a coordinate to a Haversine site resolver that decides whether the visitor is at Park Güell, the Sagrada Família, Casa Milà or Casa Batlló, and the LCD switches to that monument’s tilemap and landmark set. It takes no part in recognition and no part in the answer: the element is identified from the photograph alone, and the knowledge lookup is keyed on the classifier’s element ID. With no GPS fix the guide still works end to end — only the map defaults to the configured location instead of following the visitor.

The nine steps in detail:

1. **Choose a personality.** Artistic, Technical or Child, on the Modulino buttons. Sets the SLM system prompt and the TTS voice.
2. **Frame.** In camera mode (switch D6 high), the live view is streamed to the LCD as Base64 RGB565 chunks over Bridge RPC.
3. **Capture.** Button D7 takes a 1080p photo.
4. **Validate.** The per-monument ONNX classifier decides *which architectural element* is in frame, or rejects the photo as off-distribution and asks for a retake (§2.5).
5. **Ask.** The visitor flips to map mode and records a spoken question. Recording ends by itself on silence.
6. **Transcribe.** faster-whisper, biased towards the domain vocabulary (“Gaudí”, “trencadís”, “Sagrada Família”, ...) through an `initial_prompt`.

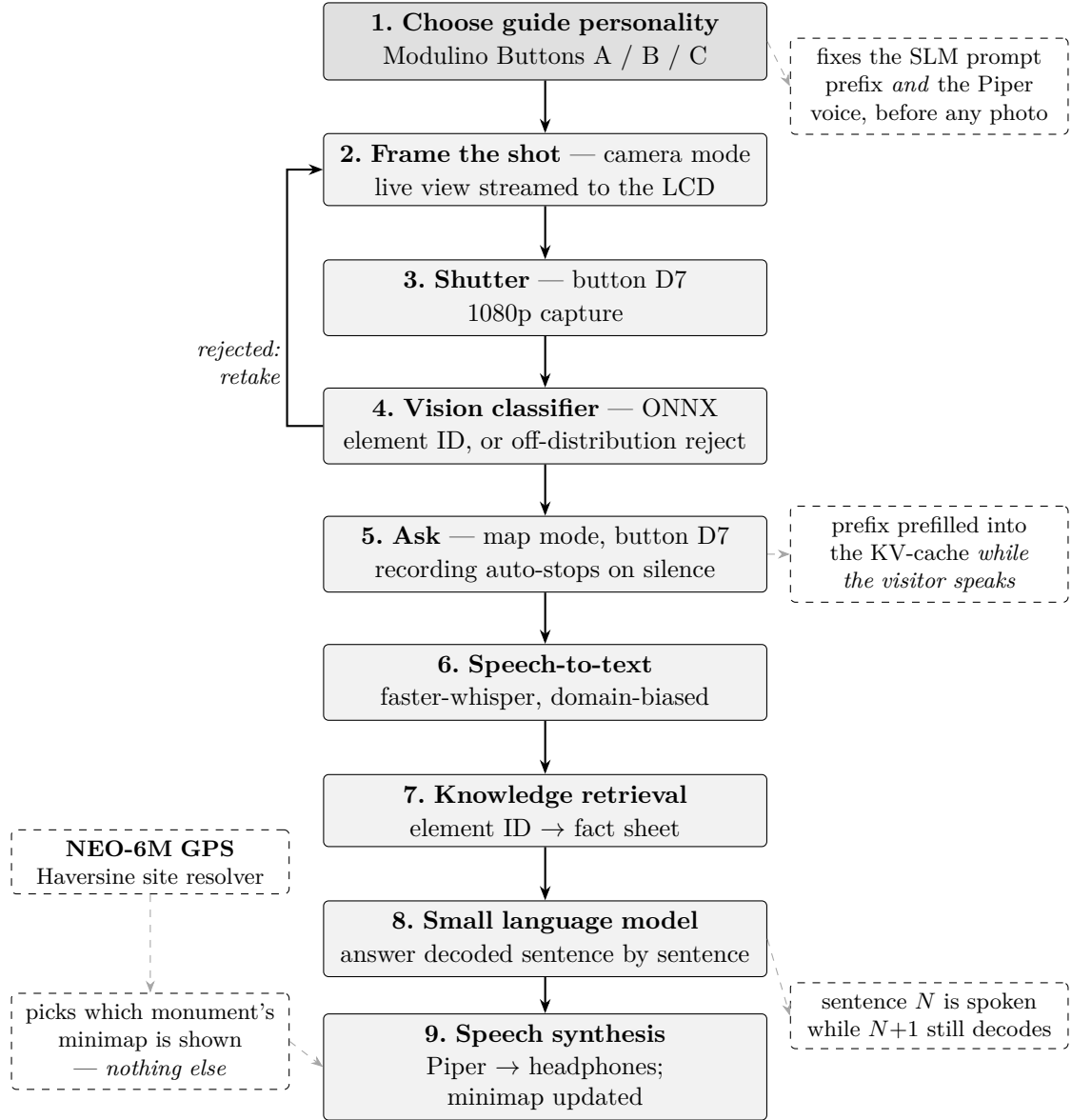


Figure 1: The Cultura Viva interaction. Solid arrows are the visitor-facing sequence; dashed arrows are influences and overlaps rather than steps. The personality is chosen once, up front (step 1), and the GPS only ever selects which minimap is displayed.

7. **Ground.** The recognised element ID indexes directly into a curated knowledge base (`knowledge_base.json`, `element_sheets.json`), with keyword search as fallback.
8. **Answer.** The retrieved facts, the personality prompt and the question go to the SLM, which decodes sentence by sentence.
9. **Speak.** Piper synthesises each sentence and streams it to ALSA while the SLM decodes the next. Volume tracks the Modulino knob live, and the visited landmark is marked on the minimap.

1.3 Design constraints

Every model choice in this report was made against the same budget:

Constraint	Value
Compute	4× ARM Cortex-A53 @ 2.0 GHz, CPU only
RAM for SLM weights + context	< 1.5 GB
SLM context window	2048 tokens
Target response latency	< 3 s to first audio
Connectivity at inference time	none (fully offline)

There is no GPU backend: the `llama.cpp` build on the board is CPU-only, and ONNX Runtime uses the CPU execution provider.

2 Vision models

2.1 Problem formulation

The vision stage answers “*what* is the visitor pointing at?”. It is a per-monument classification problem, not a single global one: each monument has its own set of architectural elements, so the label spaces are disjoint and a shared classification head would be meaningless. We therefore train one classifier per (monument, model) pair.

Classes are auto-detected from each monument’s own subfolders in the dataset, so adding a monument or a class requires no code change.

2.2 Dataset and splits

All images come from the `culturaviva/image_1080` Hugging Face dataset, one subfolder per monument. The split is fixed and shared by training, validation and threshold calibration:

Train / validation / test	70 % / 15 % / 15 %
Split seed	42
Input resolution	224 × 224
Normalisation	mean 0.5, std 0.5 per channel

Training and calibration both go through the same dataset loader, which guarantees that OOD thresholds are calibrated on images the model was never fit to.

2.3 Backbones tried

Two backbones were trained, one light and one heavy, to see whether the extra capacity of a transformer is worth paying for on this data.

With 4 monuments × 2 backbones, a full pass is **8 training runs**. Each run gets its own dataset split instance, model instance, output directory and Weights & Biases run, named `<monument>-<model>`, so runs never overwrite each other.

Table 1: Element classes per monument. Class counts differ freely; nothing in the pipeline assumes they match.

Monument	N	Element classes
Casa Batlló	2	pla_frontal pla_inferior
Park Güell	7	escalinata_drac sala_hipostila placa_natura pavellons_consergeria 3_viaductes turo_3_creus casa_museu
La Pedrera (Casa Milà)	3	facana_1 facana_2 facana_3
Sagrada Família	6	facana_naixement facana_passio laterals posterior torres cupula

Table 2: Vision backbones.

Name	Hugging Face checkpoint	Character
mobilenetv2	google/mobilenet_v2_1.0_224	lightweight CNN
vit	google/vit-base-patch16-224	ViT-B/16 transformer

Training hyper-parameters. These are *not* shared across runs. Each (monument, backbone) pair was tuned on its own: the monuments differ in class count and in how many images per class they have, and MobileNetV2 and ViT do not want the same learning rate, so a single global configuration would have handicapped several of the eight runs. `config.yaml` carries a default block that each monument and each model entry may override, and the effective values per run are those in Table 3.

Common to every run: the best checkpoint is selected by validation macro-F1, and evaluation reports **accuracy** and **macro-F1** — macro so that rare elements are not drowned out by frequent ones — on the held-out test split.

2.4 Results

What ships. Four exported classifiers ship with the device, one per monument, under `models/vision/<location>/`: Casa Batlló (2 element classes), Park Güell (7), La Pedrera (3) and the Sagrada Família (6), each plus an **unknown** label for rejected inputs. Each is a `model.onnx` plus a `labels.json` carrying the label map, input size and normalisation constants, so the board-side code needs no per-model configuration — switching monument is a directory change, nothing more.

2.5 Out-of-distribution rejection

A real visitor will point the camera at a tree, another tourist, or the sky. A plain **argmax** classifier is obliged to name an architectural element anyway, which would produce a confident, grounded,

Table 3: Effective training hyper-parameters per run. *To be filled from the corresponding Weights & Biases run configs.*

Monument	Backbone	Epochs	Batch size	Learning rate
Casa Batlló	MobileNetV2	[TBD]	[TBD]	[TBD]
Casa Batlló	ViT-B/16	[TBD]	[TBD]	[TBD]
Park Güell	MobileNetV2	[TBD]	[TBD]	[TBD]
Park Güell	ViT-B/16	[TBD]	[TBD]	[TBD]
La Pedrera	MobileNetV2	[TBD]	[TBD]	[TBD]
La Pedrera	ViT-B/16	[TBD]	[TBD]	[TBD]
Sagrada Família	MobileNetV2	[TBD]	[TBD]	[TBD]
Sagrada Família	ViT-B/16	[TBD]	[TBD]	[TBD]

Table 4: Test-split accuracy and macro-F1 per (monument, backbone) run. *These values are logged to the Weights & Biases project cultura-viva and were not exported into the repository; the placeholders below are to be filled from the corresponding W&B runs.*

Monument	N	MobileNetV2		ViT-B/16	
		Acc.	macro-F1	Acc.	macro-F1
Casa Batlló	2	[TBD]	[TBD]	[TBD]	[TBD]
Park Güell	7	[TBD]	[TBD]	[TBD]	[TBD]
La Pedrera (Casa Milà)	3	[TBD]	[TBD]	[TBD]	[TBD]
Sagrada Família	6	[TBD]	[TBD]	[TBD]	[TBD]
Mean		[TBD]	[TBD]	[TBD]	[TBD]

entirely wrong audio guide answer. The pipeline therefore puts a gate in front of the prediction. The input is rejected as `not_sure` when *either* signal fires:

$$H(p) = - \sum_{i=1}^N p_i \ln p_i > \tau_H \quad (\text{predictive entropy too high}) \quad (1)$$

$$\max_i p_i < \tau_{\text{conf}} \quad (\text{peak confidence too low}) \quad (2)$$

with $\tau_H = 0.5 \ln N$, auto-scaled to each monument’s class count so that a 2-class and a 7-class head are held to comparable standards.

What actually runs on the board. A fixed $\tau_{\text{conf}} = 0.50$, the same value for all four monuments and both backbones. Per-checkpoint calibration was deliberately not deployed.

`calibrate_thresholds.py` exists and does the fuller job — it sweeps the validation split and suggests a $(\tau_H, \tau_{\text{conf}})$ pair for a given checkpoint subject to a recall target (default: preserve 95 % of in-distribution recall), printing a ready-to-paste YAML snippet, and `inference.py` will merge overrides across three levels (*global* $\rightarrow$ *monument* $\rightarrow$ *model*, most specific winning). None of that is in use on the device. A flat 50 % confidence cut proved good enough in practice: the failure it has to catch is a photo of foliage or a passer-by, which lands nowhere near 50 % confidence on any of the heads, so the extra precision bought by per-checkpoint tuning would not have changed which photos get rejected. Calibration remains available if a future monument turns out to need it.

2.6 Export path

1. `train_classifier.py` — fine-tunes every (monument, model) combination in `config.yaml`.
2. `export_to_onnx.py` — converts one fine-tuned checkpoint into `model.onnx` + `labels.json`.
3. `calibrate_thresholds.py` — suggests per-checkpoint OOD thresholds. Available, but not used by the shipped configuration (§2.5).
4. `inference.py` — runs an exported checkpoint on a single image with the gate applied; also importable as `MonumentClassifier`, which is how the device uses it.

3 Benchmarks

Three separate benchmark suites were built, one per non-vision stage. All three target the same hardware; where a measurement was taken on a host PC rather than on the board, this is stated explicitly.

3.1 Small language model

3.1.1 Candidates

Five instruction-tuned small models were evaluated, all at Q4_K_M quantisation so that weights plus a 2k context fit the < 1.5 GB budget.

Table 5: SLM candidates. RAM footprint is weights plus 2k context.

Model	Ollama tag	Weights	RAM
Qwen 2.5 1.5B Instruct	qwen2.5:1.5b	1.10 GB	1.45 GB
Llama 3.2 1B Instruct	llama3.2:1b	0.80 GB	1.10 GB
Gemma 3 1B IT	gemma3:1b	0.85 GB	1.15 GB
SmolLM2 1.7B Instruct	smollm2:1.7b	1.15 GB	1.50 GB
Qwen 2.5 0.5B Instruct	qwen2.5:0.5b	0.39 GB	0.65 GB

3.1.2 Protocol

- **Test set.** 38 curated Gaudí questions with reference answers and retrieved contexts (`eval/testset.json`).
- **Generation.** `temperature` = 0.1, `max_tokens` = 60, `repeat_penalty` = 1.1, stop sequences `"\n\n"` and `<|im_end|>`, context window 2048, 4 threads, batch size 256. These deliberately mirror what the device passes to `create_chat_completion`: a benchmark run with a different token budget or stop condition stops describing what actually ships.
- **Retrieval.** Top-3 passages from the knowledge base, scored semantically. This differs from the device, which looks an element up directly by the ID the vision stage gives it — the test set carries no element IDs and includes architect-level and cross-monument questions, so semantic retrieval is the only option here.
- **Personalities.** Each candidate is run under all three guide voices (Artistic, Technical, Child), producing a separate predictions file per personality.
- **Judge.** Ragas, scoring *Faithfulness*, *AnswerRelevancy*, *ContextPrecision* and *ContextRecall*, driven by a *local* Ollama judge (`qwen2.5:7b`, temperature 0) with `all-MiniLM-L6-v2`

sentence embeddings. Keeping the judge local means the evaluation is reproducible offline and no test data leaves the machine.

- **On-device re-run.** `benchmark_arduino.py` re-runs the same 38 questions on the Cortex-A53 through `llama-cpp-python` built with `-march=armv8-a -mtune=cortex-a53`, and the resulting predictions are scored with the same Ragas judge back on the PC.

3.1.3 Results

Table 6: Per-answer generation wall-clock over the 38-question test set, as recorded in the checked-in prediction files. These runs were produced through Ollama on the development host; they are *not* Cortex-A53 timings and should be read as a relative ordering between candidates, not as device latency.

Model	Mean (s)	Median (s)	Min (s)	Max (s)
Llama 3.2 1B Instruct	2.42	2.40	2.10	4.33
Qwen 2.5 0.5B Instruct	2.43	2.41	2.09	3.59
Gemma 3 1B IT	2.57	2.48	2.17	5.52
Qwen 2.5 1.5B Instruct	2.61	2.61	2.14	4.97
SmolLM2 1.7B Instruct	2.79	2.65	2.18	5.96

The spread in generation time is small — roughly 15 % between the fastest and the slowest candidate — which is the important negative result: on this workload, *latency does not discriminate between the candidates*, so the choice can be made on answer quality and memory headroom instead.

Table 7: Ragas scores over the 38-question test set. *The scored results (`eval/results/`) are not checked into the repository; re-run `main.py benchmark` to regenerate.*

Model	Faithful.	Ans. Rel.	Ctx. Prec.	Ctx. Rec.
Qwen 2.5 1.5B Instruct	[TBD]	[TBD]	[TBD]	[TBD]
Llama 3.2 1B Instruct	[TBD]	[TBD]	[TBD]	[TBD]
Gemma 3 1B IT	[TBD]	[TBD]	[TBD]	[TBD]
SmolLM2 1.7B Instruct	[TBD]	[TBD]	[TBD]	[TBD]
Qwen 2.5 0.5B Instruct	[TBD]	[TBD]	[TBD]	[TBD]

3.1.4 Guide personalities

The guide personality is the one setting a visitor chooses for themselves, so it is worth asking whether it changes anything. It is implemented purely as a system-prompt tail: `get_kg_context` is personality-independent, so all three guides receive byte-identical facts and differ only in a short block of instructions.

Protocol. 24 open questions, one per element sheet, each carrying an `element_id` so retrieval takes the direct-lookup path the device uses when vision names an element. Every question is answered by all three guides at the shipped sampling settings. A judge from a *different* model family, `llama3.1:8b`, then reads each answer blind and names the guide that wrote it, choosing between paraphrased descriptions rather than the prompt text, with the options shuffled per item. Chance is therefore 33.3 %. The same judge is given three hand-written, deliberately distinct texts as a self-test on every run; it scored 3/3 throughout, so a null result is a statement about the

guides and not about the instrument. The benchmark reads the device’s own knowledge files and imports its prompt builders directly, rather than keeping copies, so these figures describe what the board actually serves.

Tone instructions did not work. The original prompts described how each guide should sound — “you speak with passion and use evocative metaphors”, “you are precise, rigorous”, “you use simple analogies and an animated tone”. On `qwen2.5:0.5b` the judge recovered the intended guide in **28 of 72** answers, 38.9 % against a 33.3 % baseline (exact binomial $p = 0.19$): indistinguishable from guessing. The child guide was identified in 4 % of its own answers. Asked why Casa Batlló’s facade resembles bones, it replied that this “gives it a sense of solidity and strength”.

Format requirements replaced them. Each guide now demands one concrete, mechanically checkable thing, and the three are deliberately orthogonal so that a guide which obeys is distinguishable from one which does not:

- **artistic** — one comparison to nature, written as “like ...”;
- **technical** — at least one number written in digits, taken from the facts supplied;
- **child** — a closing question, ending in a question mark.

Table 8: Blind identification and format compliance under the format-requirement prompts. Chance is 33.3 %. Compliance is the share of answers satisfying that guide’s stated requirement.

Model	Identification		Compliance		
	Correct	Accuracy	art.	tech.	child
Qwen 2.5 0.5B (shipped)	30/72	41.7 %	21 %	8 %	0 %
Qwen 2.5 1.5B	53/72	73.6 %	92 %	33 %	21 %

The result depends almost entirely on model size. On the **0.5B model the change is not significant**: 41.7 % against 33.3 % chance, $p = 0.09$, and the child guide still never once ended with a question across 24 answers. Three further wordings were tried — dropping the competing “answer only what the user asks” sentence, adding a worked example, and moving the requirement to the front of the prompt — and compliance never rose above 10 %. The 0.5B model does not act on the personality tail.

On the **1.5B model the same prompts work**: 73.6 % identification, $p < 0.001$, with per-guide recall of 67 %, 79 % and 75 % — the child guide rising from 4 % under tone instructions to 75 %. Median text similarity between two guides answering the same question falls from 0.62 to 0.27.

Table 9: Blind identification on `qwen2.5:1.5b`. Rows are the guide that wrote the answer, columns the judge’s choice.

true \ predicted	artistic	technical	child
artistic	16	6	2
technical	5	19	0
child	6	0	18

Consequence for the selection. The 0.5B model was chosen for RAM headroom, and this is the measured cost of that choice: the guide personality is a feature the shipped configuration does not deliver. It is not a prompt-wording problem — the identical prompts produce 74% identification one model size up. Either the personality is presented as cosmetic, or the 1.5B model is reinstated and the 1.45 GB working set budgeted for.

Grounding is the larger problem. The same judge was asked, for each answer, whether every claim was supported by the facts the model had been given — reference-free, so that style is never scored as inaccuracy.

Table 10: Answers containing at least one claim unsupported by the retrieved facts.

Guide	0.5B	1.5B
artistic	10/24 (42 %)	7/24 (29 %)
technical	7/24 (29 %)	2/24 (8 %)
child	9/24 (38 %)	15/24 (62 %)

Between a twelfth and two thirds of answers contain something the sheets do not support, depending on guide and model size.

How far to trust that. All 72 answers from the 0.5B run were then adjudicated by hand against the sheet each was given (`results/manual_audit_qwen2.5_0.5b.json`). The manual count is **23/72 (31.9%)** against the judge’s **26/72 (36.1%)** — but the closeness is partly cancellation. Per-answer agreement is 55/72, Cohen’s $\kappa = 0.48$, with 10 judge false positives and 7 false negatives. The false positives cluster on answers that are faithful but ungrammatical: one lifts 65% of its words verbatim from the sheet and was still marked unsupported, suggesting the judge partly conflates incoherence with invention. The false negatives are quiet fabrications it waved through — a date absent from the sheet, and a facade described as polychrome where the sheet says only that Gaudí *wanted* it polychromed.

So roughly a third is the right order of magnitude, but a single judge verdict should not be used to triage an individual answer. The failures that matter are flat contradictions: Casa Batlló placed “in the centre of Barcelona’s Gothic Quarter” (it is on Passeig de Gràcia), Casa Milà called “part of Gaudí’s famous Park Güell complex”, and the Hypostyle Room built “to house a collection of over 1,000 columns” where the sheet says Room of a Hundred Columns and describes a rainwater collection system.

Note also that the guide asked to speak to a child hallucinates most at 1.5B — simplifying and staying grounded pull against each other. On the evidence here this matters more than which voice delivers the answer, and it is the next item this benchmark should pursue.

A defect found during the audit, and fixed. Three answers returned the internal element identifier to the visitor verbatim — “You are seeing laterals_sagrada_familia in the photos.” Vision emits Catalan snake_case class labels, and `build_facts_block` put the label straight into the prompt, so the model read it as a proper name. In the worst case it invented around one: all three guides answered that “Gaudí built the Sala Hipostila to house a collection of columns”, a name absent from the sheet, which calls it the Hypostyle Room.

The sheets already carry English names and list the vision labels among their aliases, so the mapping existed and was simply never used. `ModelRegistry.display_name` now resolves it and the prompt reads “[Detected element in photo: Side Aisles and Lateral Naves]”. All 18 vision classes across the four monuments resolve; an element with no sheet falls back to the de-underscored identifier.

It is a correctness fix, not a grounding fix. Re-running the 0.5B study afterwards, the judge scored 20/72 unsupported against 26/72 before. That looks like an improvement and is not one: paired over the same 72 questions, 16 answers moved to supported and 10 moved the other way, an exact McNemar $p = 1.00$. The identifier no longer reaches the visitor, which is worth having on its own, but hallucination did not measurably fall.

The churn is the more useful number. Sixteen verdicts flipping in each direction on a re-run, with $\kappa = 0.48$ between judge and hand audit, puts the noise floor of this measurement near ± 6 answers. At $n = 72$ only effects larger than roughly eight percentage points are detectable at all, which is the bar any future grounding work has to clear before its result means anything.

Regenerate any figure in this section with `uv run python main.py personality -model <tag>`.

Selected. Qwen 2.5 0.5B Instruct, Q4_K_M (qwen2.5-0.5b-instruct-q4_k_m.gguf).

Given that generation latency did not separate the candidates, the decision came down to what the model costs the rest of the system. At 0.65 GB for weights plus a 2k context, the 0.5B model is less than half the working set of the 1.5B one (1.45 GB) and under half of SmolLM2 (1.50 GB). That difference is not academic: the SLM is not alone on the board. The vision classifier, faster-whisper (397 MB peak, §3.2), Piper and the persisted prefix caches all have to be resident at the same time, on a device that is also driving a camera, a display and audio. Choosing the smallest candidate that answers the questions acceptably is what keeps the pipeline from contending for memory mid-interaction — and it shortens the cold prefill that `prewarm_cache.py` has to precompute for all 24 elements.

The answer quality that “acceptably” refers to is the Ragas comparison above, which still has to be regenerated and filled in.

3.2 Speech-to-text

3.2.1 Engine

One engine was benchmarked and shipped: **faster-whisper** on the Whisper `base.en` model, running under CTranslate2 with INT8 quantisation.

`base.en` is the smallest Whisper tier that holds up on full sentences, and CTranslate2’s INT8 path is what makes it viable on four Cortex-A53 cores: it keeps the 141 MB model and its activations well inside LPDDR4 and maximises memory bandwidth, which is the binding constraint here rather than arithmetic. The English-only (`.en`) variant is deliberate — the guide answers in English, and the monolingual weights are both smaller and more accurate on English than the multilingual ones at the same size.

3.2.2 Metrics and optimisations

The harness measures **WER**, **CER**, **keyword-spotting accuracy**, **real-time factor** (inference time / audio duration) and **peak RAM**.

Keyword-spotting accuracy deserves a note. A generic WER number hides the failure that actually matters for an audio guide: a transcript that gets “Gaudí”, “trencadís” or “Sagrada Família” wrong will retrieve the wrong knowledge sheet, and the whole answer is then confidently off-topic. The decoder is therefore biased towards the domain vocabulary through an `initial_prompt`:

Cultura Viva audio guide in Barcelona about Antoni Gaudí, Sagrada Família basilica, Nativity, Passion, and Glory facades, Catalan modernisme architecture, Casa Batlló, Casa Milà, Park Güell, dragon and salamander sculptures, and trencadís mosaics.

For **faster-whisper** the same terms are additionally passed as explicit `hotwords`. Further tuning:

- **4 CPU threads** — the QRB2210 has exactly 4 Cortex-A53 cores, so 4 saturates the CPU without paying context-switch overhead.
- **INT8 compute type** — keeps the model comfortably inside LPDDR4 and maximises memory bandwidth.
- **Greedy decoding** — `temperature = 0.0`, `beam_size = 1`, to minimise RTF.

3.2.3 Results

The evaluation set is **24 synthetic visitor utterances**, varied across register (neutral, polite, child-like), intent (materials, history, ...) and speaker voice, with controlled background noise, each annotated with its target keywords.

Table 11: STT results over the 24-utterance set, domain bias applied, from the `results_computer/` run.

Engine	WER	CER	Keyword acc.	Peak RAM	Model
<code>faster-whisper:base.en</code>	0.164	0.093	96.1 %	397 MB	141 MB

A 16.4 % word error rate alongside 96.1 % keyword accuracy is exactly the shape we were optimising for: the transcript is imperfect in its function words but almost never wrong about the proper nouns that drive retrieval.

Selected. `faster-whisper` with the `base.en` model at INT8, which is what the device ships (`models/stt/faster-whisper-base.en`).

3.3 Text-to-speech

3.3.1 Candidates

18 ONNX voices are registered in `models_config.json`: three VITS models exported from Hugging Face (MMS-TTS English, VITS LJSpeech, VITS VCTK) and 15 Piper voices — Lessac, Ryan, Amy, Alan and Cori, each in low, medium and high quality. 15 of the 18 were actually timed; `piper-amy-high`, `piper-alan-high` and `piper-cori-low` were not run.

The field narrowed to Piper, on the inference-time / quality trade-off. The VITS exports were included as the obvious general-purpose baseline and were eliminated by the numbers below: at RTF ≈ 3.8 they are far too slow for a device that has to answer while the visitor is standing in front of the building, and the naturalness they return for that cost is not better than a Piper medium voice. Piper, built for exactly this class of hardware, reaches acceptable voice quality at RTF well under 1 — and, crucially, is tiered, so quality can be traded against compute by swapping a voice file rather than changing the runtime. That tiering is what the rest of this section measures.

Audio quality is not scored here — the benchmark measures *inference time and footprint only*. Which specific Piper voice suits which guide personality was decided separately, by ear.

3.3.2 Protocol

Measured on the UNO Q itself (Qualcomm QRB2210, Cortex-A53, 4 threads, 3667 MB RAM detected), 2026-08-26. Each model is loaded once, given 1 untimed warm-up run, then timed over **9 runs** round-robin balanced across sentence-length buckets — short (≤ 8 words), medium (9–16), long (≥ 17) — drawn from a fixed set of Gaudí sentences.

Table 12: TTS inference time on the Arduino UNO Q. Median seconds per sentence by length bucket; RTF is the median of (inference time / generated audio duration), so $RTF < 1$ means the voice synthesises faster than real time. Disk size is the configured on-disk model size.

Voice	Disk (MB)	Short (s)	Medium (s)	Long (s)	RTF
<i>VITS (ONNX Runtime, CPU)</i>					
MMS-TTS (English, VITS, ONNX)	138	12.91	21.95	40.92	3.88
VITS (LJSpeech, ONNX)	332	15.39	43.89	59.04	3.78
VITS (VCTK, multi-speaker, ONNX)	150	13.69	34.43	49.50	3.80
<i>Piper</i>					
Piper (en_US-lessac-low)	20	1.61	4.12	6.00	0.62
Piper (en_US-lessac-medium)	63	1.91	4.91	7.28	0.77
Piper (en_US-lessac-high)	113	12.86	33.82	50.45	5.19
Piper (en_US-ryan-low)	20	1.57	3.89	5.87	0.63
Piper (en_US-ryan-medium)	63	1.94	5.36	7.43	0.80
Piper (en_US-ryan-high)	113	12.52	37.24	51.76	5.31
Piper (en_US-amy-low)	20	1.79	4.28	5.95	0.61
Piper (en_US-amy-medium)	63	2.44	5.83	8.23	0.80
Piper (en_GB-alan-low)	20	1.85	5.07	7.30	0.64
Piper (en_GB-alan-medium)	63	2.39	6.14	8.67	0.75
Piper (en_GB-cori-medium)	63	2.27	4.51	7.19	0.81
Piper (en_GB-cori-high)	113	13.31	31.94	52.32	5.27

3.3.3 Reading the results

Table 12 separates the field cleanly into three groups.

- **The VITS exports are unusable here.** $RTF \approx 3.8$ means roughly four seconds of compute per second of speech. A 20-second answer would take over a minute to synthesise.
- **Piper “high” voices are equally unusable**, and for the same reason — $RTF \approx 5.2\text{--}5.3$, *worse* than the VITS models despite Piper’s much lighter runtime. The quality tier, not the framework, is what blows the budget.
- **Piper low and medium clear the bar with room to spare.** Every low voice lands at $RTF \approx 0.61\text{--}0.64$ and every medium voice at $\approx 0.75\text{--}0.81$. A medium voice synthesises a short sentence in about 2s and a long one in about 8s, while producing more audio than that in playback time — which is what makes sentence-level streaming (§3.4) work: playback of sentence N outlasts synthesis of sentence $N + 1$, so the audio never stalls.

The gap between low and medium is around 25 % of compute for a clearly better voice, which is worth paying given the headroom.

Selected. Two Piper medium voices ship on the device, mapped to guide personalities:

Personality	Style	Voice
A — Artistic	Evocative, metaphorical	en_US-libritts_r-medium
B — Technical	Architectural, precise	en_GB-semaine-medium (Spike)
C — Child	Simple, fun, story-led	en_GB-semaine-medium (Prudence)

3.4 End-to-end latency

`python/benchmark.py` measures the stages on the device. Its headline number is deliberately *not* the sum of them.

Adding up every stage measured in isolation gives an upper bound, not a latency, because the pipeline does not run them in sequence: vision happens at photo time, prefill overlaps with recording, and Piper synthesises while `llama.cpp` is still decoding. The benchmark therefore reports **time to first audio** — from the moment the visitor stops speaking to the first word they hear — counting only the stages still on that path, each with the figure that applies there: STT in full, the *warm* prefill rather than the cold one, decode only as far as the first complete sentence, and one sentence of Piper rather than the whole answer. It excludes the `aplay` spawn and Bridge RPC round-trips, so the board is somewhat slower than the printed figure.

Three mechanisms create that overlap:

1. **Asynchronous prefill.** Once a photo is confirmed, the static prefix — personality instructions plus the element’s fact sheet — is prefilled into the KV-cache on a background thread while the visitor is still speaking, so only the question needs prefilling at the end. The prefix cache is also persisted to disk: `prewarm_cache.py` computes it once per element (≈ 12 minutes for all 24), so no visitor ever waits through a cold prefill.
2. **Silence detection with cropping.** Recording stops by itself after ≈ 1.5 s of silence, and the silence is cropped from both ends before Whisper sees it. This is not cosmetic: the encoder pads every window to a fixed length, so room tone costs exactly as much to encode as speech. Thresholds adapt to the noise floor measured at the start of each recording.
3. **Sentence-level streaming.** Tokens are emitted sentence by sentence rather than after the full 60+-token generation, so Piper synthesises sentence N while the SLM decodes sentence $N + 1$. At the Piper medium RTF of ≈ 0.8 measured above, synthesis stays ahead of playback and the audio never stalls.

Together these turn the perceived latency from “sum of four stages” into “STT + warm prefill + one sentence of decode + one sentence of Piper”.

4 Bill of materials

4.1 Arduino components

Prices are Arduino Store (EU) list prices, VAT included, retrieved 13 September 2026 from <https://store.arduino.cc>.

Table 13: Arduino components. The three Modulino modules are the ones this project requires.

Component	Code	Qty	Role in Cultura Viva	Price
Arduino UNO Q 4GB	—	1	MCU + Linux MPU, the whole system	82.90 €
<i>Modulino modules (I²C, Qwiic / Wire1)</i>				
Modulino Buttons	ABX00110	1	Guide personality select A/B/C + LEDs	8.50 €
Modulino Knob	ABX00107	1	Live headphone volume 0–100 %	9.00 €
Modulino Buzzer	ABX00108	1	Shutter click, confirmations, alerts	7.30 €
Modulino subtotal				24.80 €
Arduino subtotal				107.70 €

4.2 Third-party components

Two of these are not optional extras but consequences of the board’s form factor, and are easy to forget when budgeting:

- **USB-C hub.** The UNO Q exposes a single USB-C port on the Linux side, and the guide needs three things on it at once — the camera, the headphones and power. A powered USB-C hub with pass-through power delivery is therefore required, not a convenience. It must carry USB-C PD pass-through (so the board keeps drawing full power while the hub is populated), at least one USB-A port for the webcam, and a 3.5 mm audio output or a USB audio path for the headphones.
- **Power bank.** The guide is carried around a monument, so it runs on battery, and it must run at the board’s *high-performance* profile — four Cortex-A53 cores saturated by STT, SLM decode and Piper, with a USB camera streaming at the same time. The power bank therefore needs USB-C Power Delivery at 5 V/3 A (≥ 15 W) sustained, not a 5 W phone charger which will brown the board out mid-answer. Capacity $\geq 10\,000$ mA h covers a full visit with the display and radios on.

Table 14: Non-Arduino components. *These prices were not verified against a named vendor — replace with your actual purchase prices before publishing.*

Component	Interface	Qty	Role	Price
USB-C hub, PD pass-through	USB-C	1	Camera + headphones + power on the board’s single USB-C port	[TBD]
Power bank, USB-C PD ≥ 15 W, $\geq 10\,000$ mA h	USB-C	1	Battery operation at the high-performance profile	[TBD]
ST7735S LCD, 1.8 " 128×160	SPI	1	Live view, minimap, UI	[TBD]
Logitech Brio 105 webcam	USB 2.0	1	1080p capture + microphone	[TBD]
NEO-6M GPS module	Serial1 @ 9600	1	Selects the active minimap	[TBD]
Mode switch (SPDT)	Digital D6	1	Camera / map mode	[TBD]
Push button	Digital D7	1	Shutter / record / cancel	[TBD]
Headphones or speaker	3.5 mm jack	1	Spoken answer output	[TBD]
Jumper wires, carrier board	—	—	Assembly	[TBD]
Third-party subtotal				[TBD]

4.3 Software cost

Zero. Every model in the pipeline is open-weights and runs locally, so there are no inference API calls and no per-visitor marginal cost. The Ragas judge used during evaluation (qwen2.5:7b via Ollama) is also local, and is a development-time dependency that never ships to the device.

5 Summary of selections

6 Open items

The following are referenced above but not currently exported into the repository, and appear as [TBD]:

Table 15: What the benchmarks selected, and on what grounds.

Stage	Selected	Decided on
Vision	Per-monument ONNX classifiers, 224×224 , with entropy + confidence OOD gate	Accuracy / macro-F1 per monument; OOD rejection is a hard requirement, not a nicety
STT	faster-whisper <code>base.en</code> , INT8, 4 threads, greedy	16.4 % WER but 96.1 % keyword accuracy — proper nouns drive retrieval
SLM	Qwen 2.5 0.5B Instruct Q4_K_M	Smallest working set (0.65 GB) that answers acceptably; latency did not discriminate between candidates, so memory headroom decided it
TTS	Piper medium voices, per personality	RTF ≈ 0.8 on the board, which is what makes sentence-level streaming viable

1. **Vision accuracy / macro-F1** for the 8 (monument, backbone) runs (Table 4), and the **per-run hyper-parameters** (Table 3). Both are in the Weights & Biases project `cultura-viva; vision/outputs/` is gitignored.
2. **Ragas scores** for the 5 SLM candidates. `eval/results/` is not checked in; regenerate with `uv run python main.py benchmark`.
3. **Third-party component prices**, including the USB-C hub and power bank.
4. **Measured time-to-first-audio** from a board run of `python/benchmark.py -json`.

TTS inference time by sentence length, vs. model size

Arduino UNO Q (Qualcomm QRB2210, Cortex-A53, Linux) · median $\pm$ min/max per length bucket, over Gaudí-related sentences

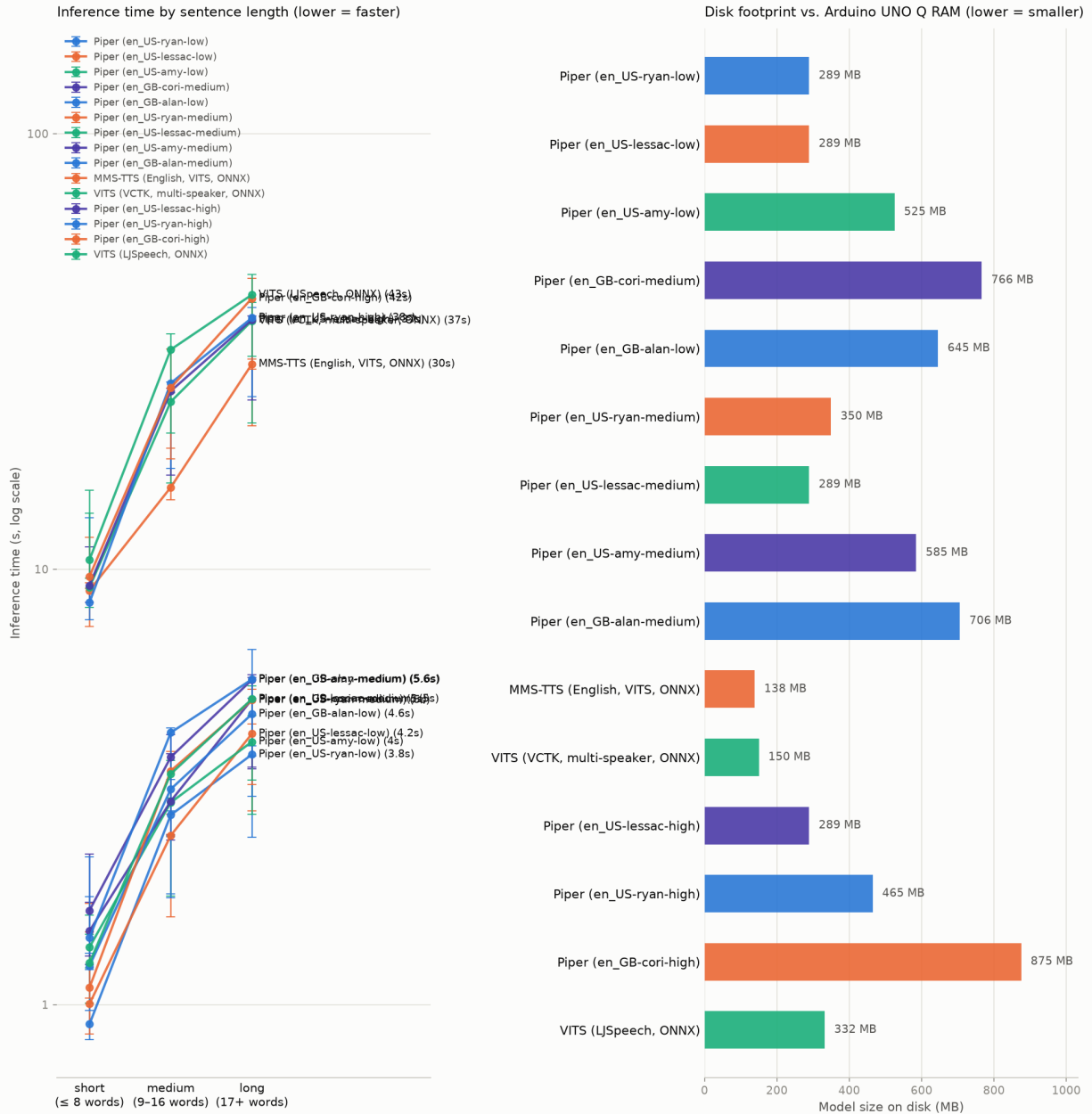


Figure 2: Median inference time with min/max whiskers per sentence-length bucket, alongside on-disk model size, generated by `visualize.py` from the same results file as Table 12.